VACATION MINI PROJECT REPORT

PASSWORD GENERATOR & VAULT

A Client-Side Password Generation and Secure Local Vault Management System

Project Report submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Computer Applications (BCA)

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CERTIFICATE

This is to certify that the mini project entitled

PASSWORD GENERATOR & VAULT

has been successfully developed and submitted by **PAVAN** (Roll No. 24BCA100), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru, in partial fulfillment of the academic requirements for the degree of Bachelor of Computer Applications.

This mini project was developed independently as part of self-learning and skill enhancement during the vacation period. The work presented in this report is original and has not been submitted elsewhere for any academic examination or award.

The project demonstrates a satisfactory understanding of front-end web technologies and the ability to design and implement a functional, security-conscious real-world application.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **PAVAN**, student of 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru (Roll No. 24BCA100), hereby declare that the mini project report entitled **"PASSWORD GENERATOR & VAULT"** submitted to the Department of BCA is a record of original work done by me during the vacation period.

I further declare that:

- This project has been carried out independently, without any formal institutional guidance, as a self-initiated learning exercise.
- The work presented in this report is genuine and has not been copied or reproduced from any other source without proper acknowledgement.
- This project has not been submitted previously for the award of any degree, diploma, or any other academic title at this or any other institution.
- All references and sources of information used in this project have been duly cited and acknowledged.
- The data presented, the analysis performed, and the conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru

Date: July 2026

Signature of the Student

PAVAN

Roll No: 24BCA100

4th Semester, BCA

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and encouraged me throughout the development of this mini project.

I am grateful to the creators and contributors of the open educational resources available through platforms such as MDN Web Docs, W3Schools, Stack Overflow, and YouTube, which served as invaluable references during development. These resources helped me overcome technical challenges and strengthen my understanding of HTML, CSS, and JavaScript.

I extend heartfelt thanks to my family for their constant encouragement and for providing a supportive environment for learning during the vacation. Their motivation kept me focused and committed throughout the project.

This project was undertaken entirely on my own initiative. Independently planning, designing, and deploying a security-oriented web application has greatly enhanced my practical skills and my appreciation of responsible software development.

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ABSTRACT

In an era where digital accounts and online services have become central to everyday life, the security of personal passwords is of paramount importance. The widespread practice of reusing simple, predictable passwords across multiple platforms exposes users to significant security risks, including data breaches, account takeovers, and identity theft. Existing password managers, while effective, often require cloud accounts, subscriptions, or browser extensions, creating barriers for users who prefer simplicity and privacy.

The **Password Generator & Vault** is a lightweight, client-side web application that addresses these concerns by providing a fully local, no-signup password generation and management solution. Users can generate cryptographically strong passwords with configurable parameters and store them in a browser-based vault — all without any data leaving their device.

Key Features:

- Configurable password generation: length (up to 50+ characters), uppercase, lowercase, numbers, symbols, and ambiguous character exclusion.
- Real-time password strength indicator (Weak / Fair / Good / Very Strong).
- One-click copy to clipboard functionality.
- Local Vault for storing passwords against named accounts and usernames.
- Vault search by account name or username.
- About page with transparent security notes and usage instructions.
- Fully offline-capable — no server, no login, no data transmission.

Technologies Used:

HTML5, CSS3, Vanilla JavaScript, Web localStorage API. Deployed on Netlify.

Expected Outcome:

The application empowers users to generate and manage strong, unique passwords for every account without relying on external services, thereby improving personal cybersecurity hygiene in a private, transparent, and accessible manner.

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CHAPTER 1

INTRODUCTION

1.1 Background

The proliferation of online services has made password management one of the most pressing personal security concerns of the modern digital age. Security researchers consistently report that the majority of data breaches involve weak, stolen, or reused passwords. Despite widespread awareness of this issue, many users continue to rely on simple, memorable passwords because generating and remembering complex ones is inconvenient.

A dedicated password generator and vault application addresses this gap by automating the creation of strong passwords and providing a secure, accessible place to store them. When such a tool operates entirely on the client side — with no server communication — it offers users the additional assurance that their credentials never leave their device.

1.2 About the Project

The **Password Generator & Vault** (branded as **vault.**) is a single-file web application developed using HTML5, CSS3, and Vanilla JavaScript. It is structured around three distinct views — Generator, Vault, and About — each accessible via a top navigation bar. The application requires no installation and is fully functional offline after the first load.

The live application is accessible at: **https://pass-generatorz.netlify.app/**, deployed via Netlify's static hosting platform.

1.3 Scope of the Project

The project is scoped to cover the following functional areas:

- Customisable, cryptographically sound password generation.
- Real-time strength evaluation and visual feedback.
- Local, browser-based credential vault with search capability.
- Transparent security documentation through the About page.
- Fully offline, privacy-first architecture using localStorage.

CHAPTER 2

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of this project:

1. To provide a reliable, configurable password generator: Allow users to generate passwords of any desired length with full control over character sets including uppercase, lowercase, digits, symbols, and exclusion of visually ambiguous characters.

2. To implement real-time password strength analysis: Evaluate and display the strength of each generated password instantly, helping users make informed decisions about password quality before saving or using it.

3. To build a local credential vault: Enable users to save account names, usernames, and generated passwords in a structured, searchable local vault stored entirely within the browser, requiring no external database.

4. To ensure complete user privacy: Design the system so that no data is transmitted to any server, API, or third-party service, giving users full control and ownership of their credentials.

5. To deliver a frictionless user experience: Eliminate barriers such as registration, login, or installation; the application must work immediately in any modern browser by simply opening a URL.

6. To promote cybersecurity awareness: Include a transparent About page that educates users about the storage mechanism, its limitations, and guidance for when a more robust solution may be appropriate.

7. To demonstrate practical front-end development competence: Apply core web technologies — HTML, CSS, JavaScript — to build a complete, production-deployed application that solves a real user problem.

CHAPTER 3

EXISTING SYSTEM

3.1 Current Approaches to Password Management

Users currently rely on a variety of approaches to generate and manage passwords, each carrying notable limitations:

a) Manual Password Creation

Most users create passwords manually, typically choosing short, memorable strings based on personal information (names, birthdays) or common words. Such passwords are highly vulnerable to dictionary attacks and brute-force cracking.

b) Cloud-Based Password Managers

Tools like LastPass, 1Password, or Bitwarden offer comprehensive password management but require account creation, internet connectivity, and — in most cases — a paid subscription. Users must also trust a third-party server with their credentials.

c) Browser Built-in Password Managers

Modern browsers offer to save and autofill passwords, but these are tied to a specific browser and account ecosystem, offer limited cross-platform flexibility, and provide no standalone password generation with strength customisation.

d) Online Password Generators

Websites that generate random passwords exist, but users cannot verify whether the generated password is transmitted to or logged by the server, raising significant privacy concerns.

3.2 Limitations of Existing Solutions

- Require internet connectivity and online accounts.
- Involve trust in third-party servers for password storage.
- Paid subscription models limit accessibility for casual users.
- Lack transparency regarding data handling and storage mechanisms.
- Do not combine generation and local storage in a single, standalone, offline-capable tool.
- Complex interfaces with steep learning curves for non-technical users.

CHAPTER 4

PROPOSED SYSTEM

4.1 Overview

The Password Generator & Vault proposes a self-contained, offline-first web application that combines password generation and local credential storage in a single, transparent tool. It is designed to be instantly accessible, completely private, and straightforward to use — without requiring any account, payment, or server communication.

4.2 Advantages Over the Existing System

- **Zero Trust Required:** All data remains on the user's device. No credentials are sent to any server.
- **No Registration or Login:** Open the URL and use it immediately — no barriers to entry.
- **Fully Offline Capable:** Works without an internet connection after the initial page load.
- **Transparent Security:** The About page explicitly documents how data is stored and recommends alternatives for high-security use cases.
- **Combined Generation and Storage:** Users can generate a password and save it to the vault in a single workflow, eliminating the need for separate tools.
- **Customisable Generation:** Granular control over password parameters gives users the flexibility to meet any platform's password policy requirements.

4.3 System Architecture

The application follows a single-page application (SPA) pattern with three navigable views rendered dynamically via JavaScript. All data persistence is handled by the browser's localStorage API. No external libraries, frameworks, or APIs are used. The application is distributed as a single HTML file that can be opened locally or served from any static hosting platform.

CHAPTER 5

MODULES AND FEATURES

The application is divided into three primary modules, accessible through the top navigation bar. Each module is described in detail below.

5.1 Password Generator Module

The Generator is the landing view of the application and its core feature. It provides a comprehensive configuration panel above a live password display.

Configuration Panel:

A slider control allows users to select password length from a minimum of 4 to a maximum of 64 characters, with the current value displayed dynamically. Four checkboxes control the character sets included in generation:

- Uppercase letters (A–Z)
- Lowercase letters (a–z)
- Numeric digits (0–9)
- Special symbols (!@#…)

An additional option allows exclusion of visually ambiguous characters (0, O, l, 1) to prevent transcription errors when passwords need to be typed manually.

Password Display and Strength Indicator:

The generated password is displayed in a monospace output field with a one-click copy-to-clipboard button. A colour-coded strength bar beneath the field evaluates the password in real time, labelling it as Weak, Fair, Good, or Very Strong based on length and character variety.

Generate and Save Controls:

The 'Generate' button creates a new password using the current configuration. The '+ Save to Vault' button opens a dialog allowing the user to associate the generated password with an account name, username, URL, and optional notes before saving it to the local vault.



Figure 5.1 – Password Generator Module with Configuration Panel and Strength Indicator

5.2 Vault Module

The Vault module serves as the application's credential management interface. All entries saved from the Generator are listed here and persist across browser sessions.

Entry Display:

Each vault entry is displayed as a card showing an account icon, the account name, and the associated username or email address. A dropdown arrow reveals the full entry details including the stored password and any notes.

Search Functionality:

A search bar at the top of the vault allows users to filter entries in real time by account name or username, making it quick to locate specific credentials even as the vault grows.

Add, Edit, and Delete:

The '+ Add' button opens the entry creation form independently of the generator. Each existing entry has edit (pencil icon) and delete (trash icon) controls for individual management. All changes are immediately persisted to localStorage.

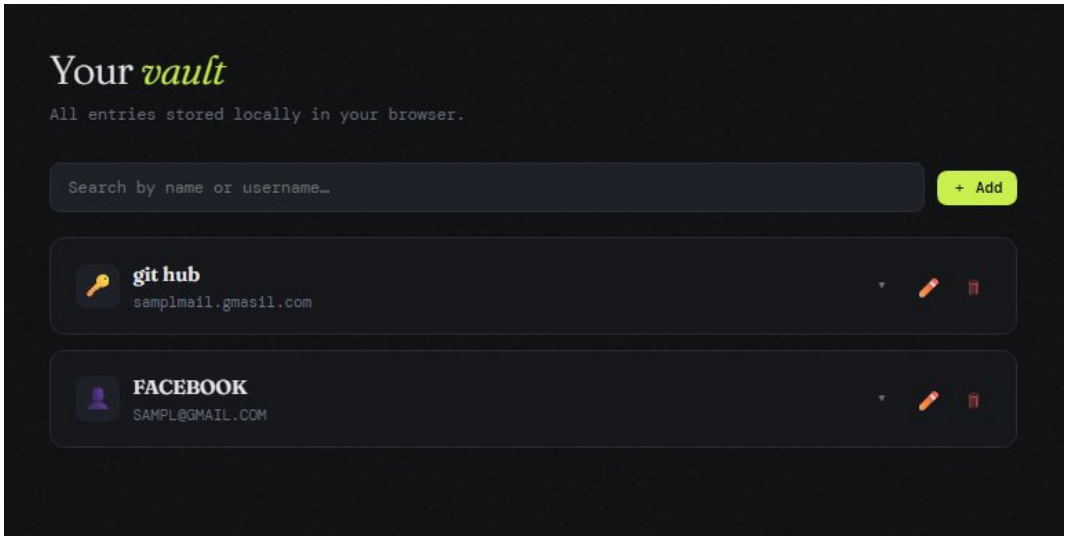


Figure 5.2 – Vault Module Showing Saved Account Entries

5.3 About Module

The About page provides essential context about how the application works and its security model, demonstrating a commitment to transparency that distinguishes this project from many proprietary tools.

How to Run:

Step-by-step instructions explain that the application can be saved as a single HTML file and opened locally in any browser, or accessed via the hosted URL. No internet connection is required after the first load.

Security Note:

A prominently displayed warning informs users that passwords are stored in plain text within the browser's localStorage. It advises against use on shared or public computers and recommends dedicated solutions such as Bitwarden or 1Password for high-security needs. This honest disclosure reflects sound, user-first design principles.

Optional Enhancements Listed:

The About page transparently lists features planned for future consideration, including AES-256 encryption, import/export functionality, TOTP/2FA storage, auto-lock, and a browser extension — demonstrating the developer's awareness of the product roadmap.

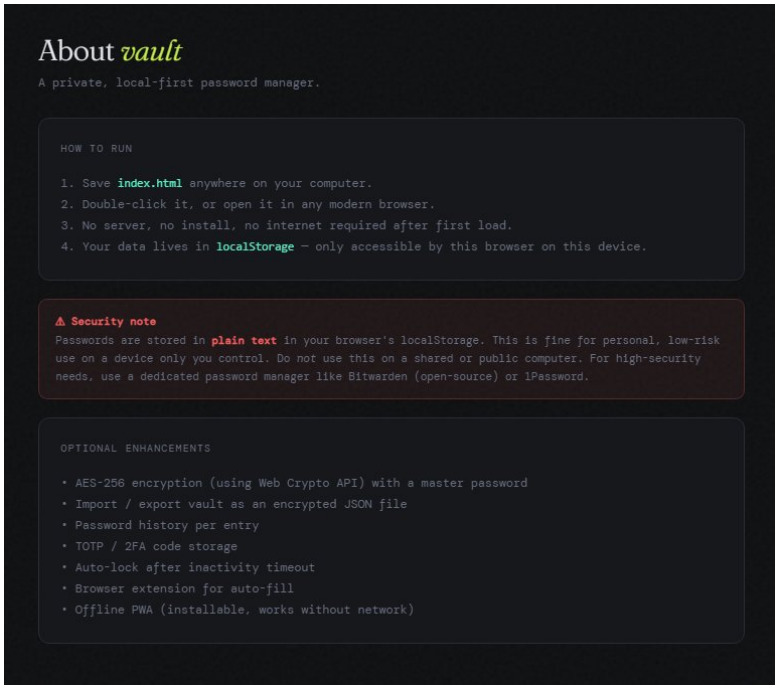


Figure 5.3 – About Module with Security Notes and Usage Instructions

CHAPTER 6

TECHNOLOGIES USED

The project was built exclusively with native front-end technologies. No frameworks, libraries, or back-end infrastructure were used. The following technologies were employed:

6.1 HTML5 (HyperText Markup Language 5)

HTML5 provides the structural foundation of the application. Semantic elements organise the three-view layout, while standard form controls (range sliders, checkboxes, text inputs) power the configuration panel and vault entry forms. HTML5's data attributes facilitate dynamic behaviour without additional markup overhead.

6.2 CSS3 (Cascading Style Sheets 3)

CSS3 delivers the application's distinctive dark-themed aesthetic. A colour palette anchored by a lime-green accent (#c6f135) against a near-black background creates high contrast and readability. CSS Custom Properties enable consistent theming, Flexbox handles responsive layouts, and CSS transitions provide smooth interactive feedback on buttons and strength indicators.

6.3 JavaScript (ES6+, Vanilla)

All application logic is implemented in Vanilla JavaScript without any external library. Key areas of implementation include: randomised password generation using character set arrays, real-time strength scoring based on entropy and character variety, dynamic DOM manipulation for view switching and vault rendering, clipboard API integration for one-click copying, and event-driven form handling for vault CRUD operations.

6.4 Web localStorage API

The localStorage API provides persistent, session-independent key-value storage within the browser. Vault entries are serialised as JSON and stored locally, allowing data to survive page refreshes and browser restarts without any server-side storage. All read and write operations are synchronous and instant.

6.5 Netlify (Static Hosting)

The application is deployed on Netlify, a leading static site hosting platform offering a global CDN, free HTTPS, and continuous deployment. The deployment process requires only uploading the single HTML file, making it an ideal choice for a client-side-only project of this nature.

#	Technology	Standard / Version	Purpose
1	HTML	HTML5	Page structure and form controls

2	CSS	CSS3	Dark theme, layout, animations
3	JavaScript	ES6+ Vanilla	Generation logic, DOM, clipboard, events
4	localStorage	Web Storage API	Client-side vault persistence
5	Netlify	Cloud Platform	Static hosting and deployment

CHAPTER 7

CONCLUSION

The Password Generator & Vault project successfully addresses a real and widely experienced problem in everyday digital life — the difficulty of creating and managing strong, unique passwords for multiple online accounts. By combining a configurable password generator, a real-time strength evaluator, and a browser-based credential vault into a single, offline-capable web application, the project delivers a practical and privacy-respecting solution accessible to anyone with a modern web browser.

From a technical standpoint, the project demonstrates that complex, security-oriented functionality can be achieved using only the core web platform — HTML5, CSS3, and Vanilla JavaScript — without relying on any external framework or back-end service. Implementing features such as character-set-based random generation, entropy-based strength scoring, clipboard API integration, and localStorage-backed CRUD operations deepened the developer's practical understanding of browser APIs and event-driven programming.

The decision to include a transparent About page with a clear security disclaimer reflects an important aspect of responsible software development — honesty with users about the limitations of a system. This is a hallmark of mature, user-centric design thinking that goes beyond mere functionality.

On a personal level, undertaking this project independently during the vacation period has been a highly rewarding experience. The end-to-end development process — from conceptualisation and UI design through to logic implementation and live deployment on Netlify — provided a comprehensive view of the front-end development lifecycle and reinforced the value of self-directed, project-based learning.

In summary, the Password Generator & Vault is a well-conceived, fully functional, and publicly accessible mini project that achieves all its stated objectives and demonstrates meaningful progress in front-end web development competence.

Live Project: https://pass-generatorz.netlify.app/